

# MODEL HIPAT5042533

## 0.5-4.2GHz Power Amplifier



Note: The photo is for illustration purposes only.  
Please refer to outline drawing

### ■ Features

- Ultra Wide Band: 0.5-4.2GHz
- Gain: 25dB
- Output Power Psat: 36dBm
- Bias: Vd=28V;Id=1.5A

### ■ Applications

- Research and Development
- Telecommunications
- Wireless Infrastructure
- TR Modules

### □ Electrical Specifications

| Parameter                                | Min.                               | Typ. | Max  | Units |
|------------------------------------------|------------------------------------|------|------|-------|
| Frequency Range                          | 0.5-4.2                            |      |      | GHz   |
| Small Signal Gain                        | 20                                 | 25   |      | dB    |
| Power Gain Flatness                      |                                    | ±1   | ±1.5 | dB    |
| Input VSWR                               |                                    | 1.5  | 2.0  | -     |
| Input VSWR                               |                                    | 2.5  | 3.3  |       |
| Output Power for 1 dB Compression (P1dB) | 31                                 | 34   |      | dBm   |
| Saturated Output Power (Psat)            | 33                                 | 36   |      | dBm   |
| Input Max Power(no damage)               |                                    |      | 20   | dBm   |
| Noise Figure                             |                                    | 10   | 15   | dB    |
| Non-Harmonic Spurious at Pout=4W         | -60                                |      |      | dBc   |
| Second and Third Harmonics at Pout=2W    |                                    | 15   |      | dBc   |
| OIP3                                     | 40                                 | 44   |      | dBc   |
| DC Supply Voltage                        |                                    | 28   | 30   | V     |
| DC Current (Vcc=+28V) (@Pout=5W)         |                                    | 1.5  | 2.5  | A     |
| Impedance                                | 50                                 |      |      | Ω     |
| Control Connector DB9                    | Pin: 1,2,3 4,5 NC; 6,7 Vd, 8,9 GND |      |      |       |
| Input Output Connector                   | SMA-k/SMA-K                        |      |      |       |
| Material                                 | Aluminum                           |      |      |       |
| Weight                                   | 1000g                              |      |      |       |
| Dimension                                | 165X132X66mm With Heatsink and Fan |      |      |       |

### Environmental Conditions

#### Environmental Specifications:

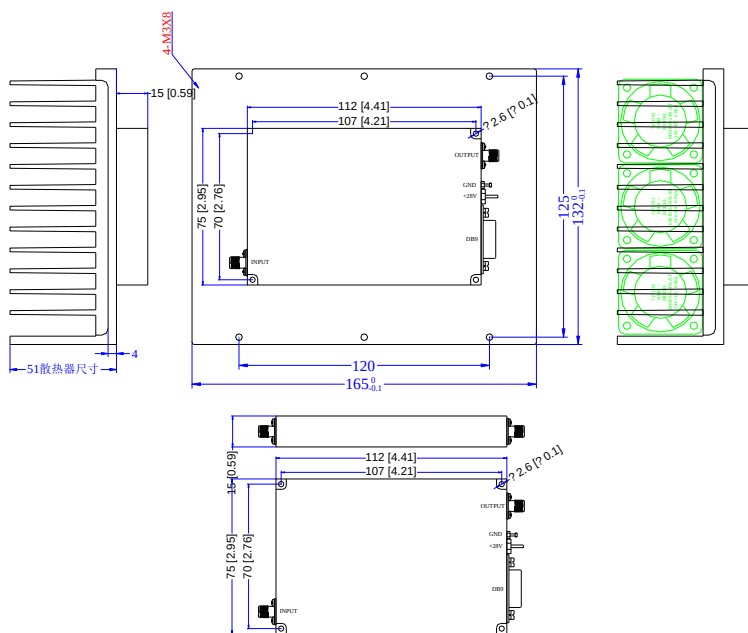
- ※ Operational Temperature       $-25^{\circ}\text{C} \sim +70^{\circ}\text{C}$
- ※ Storage Temperature             $-55^{\circ}\text{C} \sim +125^{\circ}\text{C}$



OBSERVE  
PRECAUTIONS ELECTROSTATIC  
SENSITIVE DEVICES

### Outline Drawing

All Dimensions in mm (inches) Tolerance  $\pm 0.25$  (0.01)



#### NOTE:

1. The product is designed to meet environmental ratings but not tested. If you need to test environmental condition, please contact our sales department.
2. HiMicrowave technologies co., Ltd. reserves the right to change the above information without notice.

**\*\*\*Heat Sink required during operation\*\*\***